

The Unknowns Team



CNC Pen Plotter

Teams Members

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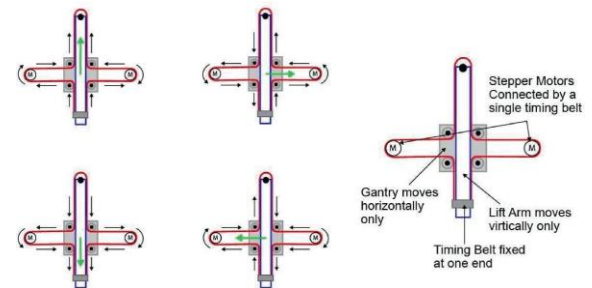
1. Project Overview:

The CNC Plotter is a computer-controlled machine designed to create precise drawings, engravings, or cuts based on a digital design input. It operates on the principle of translating digital coordinates (from a CAD file into accurate movements of a tool (such as a pen, laser, or cutter) along the X, Y, and Z axes.

This project involves designing and building a CNC Plotter using a timing belt drive mechanism powered by stepper motors, enabling the tool to move across a defined work surface for various applications such as engraving, plotting, or cutting materials like paper, wood, or plastic.

2. Working Principle of CNC Plotter:

The CNC plotter's working principle is based on the **timing belt drive mechanism**, which controls the horizontal (X-axis) and vertical (Z-axis) movements of the tool. Below is an explanation of how the CNC plotter operates:



Core Parts of the CNC Plotter:

1. Gantry (X-Axis) Movement:

- The **gantry** moves only along the **X-axis**, carrying the tool (pen or cutter) horizontally across the workspace. The horizontal motion is achieved by a **timing belt** connected to a stepper motor that drives the gantry back and forth.

2. Lift Arm (Z-Axis) Movement:

- The **lift arm** is attached to the gantry and moves only along the **Z-axis**, raising or lowering the tool for drawing or engraving. This vertical motion is also controlled by a timing belt, driven by a separate stepper motor.

3. Timing Belt (Main Movement Mechanism):

- The **timing belt** system forms a continuous loop around pulleys, with one end fixed. Motors control the belt's movement, pulling the gantry horizontally or moving the lift arm vertically.

4. Stepper Motors and Pulleys:

- **Stepper motors** drive the timing belts that control the motion of the gantry and lift arm. **Pulleys** guide the belts, converting the rotational movement of the motors into linear motion.

The timing belt system, driven by stepper motors, ensures precise and repeatable movement along the X and Z axes. By coordinating these movements, the CNC Plotter can create detailed, accurate designs based on digital inputs.

2. Materials Used and Their Mechanism:



1. Stepper Motors:

- **Function:** Provide precise control of the gantry (X-axis) and lift arm (Z-axis) movements. They rotate in precise increments (steps), enabling fine positioning of the attached parts.
- **Mechanism:** Stepper motors convert electrical pulses into discrete rotational movement. Each pulse moves the motor by a fixed angle (or step), allowing the CNC plotter to achieve accurate and repeatable movement along the axes. By adjusting the speed and number of pulses, the controller precisely controls the motion of the gantry and lift arm.
- **Quantity:** 2 (one for X-axis and one for Y-axis movement).

2. Timing Belt:

- **Function:** Transfers motion from the stepper motors to the gantry (X-axis) and lift arm (Z-axis). It forms a continuous loop and is responsible for moving parts in a precise linear direction.

- **Mechanism:** The timing belt is driven by pulleys connected to the stepper motors. As the stepper motor rotates, the timing belt moves in response, either pushing or pulling the gantry or lift arm. The belt is kept under tension to ensure consistent movement without slipping.
- **Quantity:** 1

3. Gantry Frame:

- **Function:** Supports the plotting tool and moves it horizontally across the workspace. It connects to the timing belt, which drives its movement along the X-axis.
- **Mechanism:** The gantry frame is attached to the timing belt and is driven by the stepper motor. The motor's rotation causes the belt to move, which in turn moves the gantry in a linear path along the X-axis. The gantry is mounted on linear rails or tracks to guide its movement and minimize friction.

4. Lift Arm:

- **Function:** Holds the tool (pen, cutter, etc.) and moves it vertically along the Z-axis, allowing it to engage with or lift off from the work surface.
- **Mechanism:** Similar to the gantry, the lift arm is connected to a timing belt driven by a stepper motor. The belt's movement translates into vertical motion, raising or lowering the tool. The arm is guided by vertical rails to ensure smooth and precise up-down movement.

5. Pulleys:

- **Function:** Guide the timing belts around the frame and ensure smooth, controlled movement. The pulleys transmit the rotational force from the stepper motors to the timing belts.
- **Mechanism:** Pulleys are mounted at key points to direct the path of the timing belts. As the motor turns the pulley, it pulls or pushes the belt, driving the connected gantry or lift arm. The pulleys are designed to maintain tension in the belt, ensuring accurate movement without slippage.
- **Material:** metal .

6. Frame/Structure:

- **Function:** Provides the main support for the entire CNC plotter, housing the motors, belts, pulleys, and plotting tool. It ensures stability and precision during operation.
- **Mechanism:** The frame serves as a rigid structure that holds all moving parts in place. It ensures that the gantry and lift arm move in their designated paths without wobbling or

deviations. The frame must be sturdy enough to handle vibrations while keeping all components aligned.

- **Material:** PLA 3D printed

7. **Control Board (Arduino):**

- **Function:** Acts as the "brain" of the CNC plotter, processing the digital design and sending signals to the stepper motors to control their movement along the X and Z axes.
- **Mechanism:** The control board interprets the G-code or design file and translates it into electrical signals. These signals are sent to the stepper motors, instructing them to move the timing belts in specific increments, thus controlling the movement of the gantry and lift arm. The control board also manages the speed and direction of movement.
- **Material:** Electronic components (e.g., microcontroller, circuitry).

8. **Power Supply:**

- **Function:** Provides the necessary electrical power to the control board and stepper motors.
- **Mechanism:** The power supply delivers a stable voltage and current to ensure the motors and control board function properly. It converts standard AC power into DC power, which is required by most stepper motors and control boards.
- **Material:** Electrical components (transformer, converter, etc.).

9. **Tool (Pen):**

- **Function:** The working end of the plotter, used for drawing, cutting, or engraving based on the project requirements.
- **Mechanism:** The tool is mounted on the lift arm and moves in response to the gantry and arm's movements. The tool either draws or cuts based on its vertical position (controlled by the lift arm). When the lift arm lowers the tool, it makes contact with the work surface to perform the task.
- **Material:** Dependent on application; for example, a pen for plotting or a laser for engraving.

4. Project Milestones:

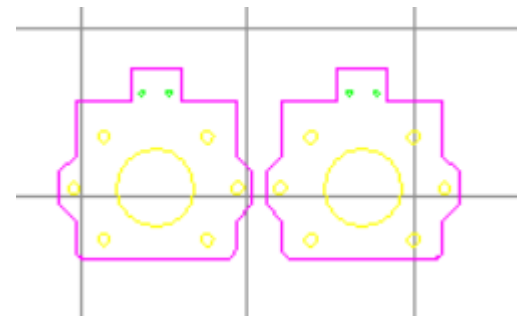
1. Design Phase (1st week):

- **Objective:** Develop a digital design for the CNC Plotter, including the frame, gantry, lift arm, and motor placements.

1. Designing Components in Fusion 360 and RDWorks

a. Stepper Motor Supporters (Designed in Fusion 360, Cut in RDWorks)

The stepper motor supporters are critical to securely hold the motors in position. They need to be designed with precision to handle the weight of the motors while ensuring stability during operation.



The initial phase of the design involved creating a detailed 3D model of the supporters using Fusion 360. In this software, exact dimensions of the stepper motors were input to ensure a perfect fit.

The design considered the necessary cutouts for ventilation, as stepper motors can generate heat during operation. By using Fusion 360's parametric design capabilities, various iterations were created, and adjustments were easily made.

Attention was paid to the thickness of the material (acrylic) which was modeled in Fusion 360, ensuring it would be sturdy enough once laser-cut.

Figure: In the figure (referenced image), the motor supports are shown with precise cutouts for screws and slots for easy attachment to the frame.

After completing the design in Fusion 360, the 3D model was exported as a 2D DXF file, which is compatible with RDWorks.

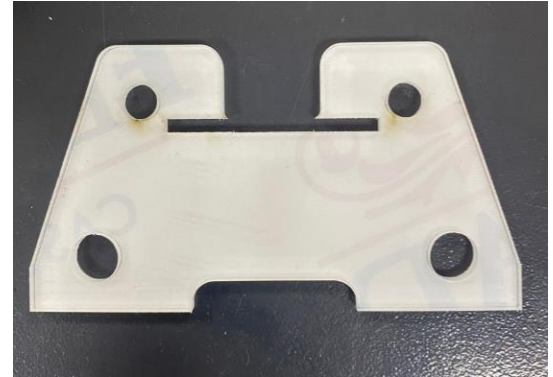
In RDWorks, the DXF file was loaded, and adjustments were made to optimize the laser cutting paths. Power settings and speed for the laser were calibrated based on the material being used to avoid burning the edges.

b. Parts Connected to Shafts (Designed in Fusion 360, Cut in RDWorks)

The components that connect the stepper motors to the shafts are essential for translating motor movement into the precise motion of the machine's axes. These parts needed to fit tightly on both the motor shafts and the machine's shafts.

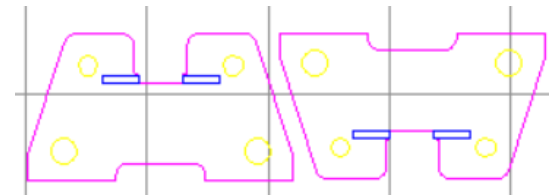
- **Design in Fusion 360:**

- Using **Fusion 360**, a parametric design approach was employed to create these couplers. The design had to accommodate the shaft diameters and ensure a secure fit, so the movement is accurately transferred without any slippage.
- Special consideration was given to the material's flexibility and durability, with slots or grooves added to the design to provide a degree of flexibility for a tight fit. The referenced figure shows the coupler designs, demonstrating how they connect the motor shafts to the rest of the mechanical system.



- **Conversion to RDWorks:**

- As with the stepper motor supporters, the 3D models were exported as DXF files and imported into **RDWorks**.
- The laser cutting parameters were again adjusted, and the toolpaths were optimized for smooth cutting, taking into account the material thickness.
- The final components were cut and tested for fit on both the motor shafts and the machine's axes.



- **Challenges:**

- A key challenge was ensuring that the laser-cut parts fit perfectly on the shafts without being too tight or too loose.

c. Pen Holder (Designed in Fusion 360, 3D Printed)

The pen holder is the critical component that needs to securely hold a pen and allow for smooth up and down movements. Unlike the other parts, which were laser-cut, the pen holder was 3D printed for added precision and customizability.

- **Design in Fusion 360:**

- In **Fusion 360**, the pen holder was designed with a customizable slot to fit different pen sizes. A clamping mechanism was incorporated to allow for easy pen changes without needing to dismantle the holder.
- The design also accounted for the weight and balance of the pen to ensure it would remain upright during the CNC machine's operation. The figure shows the 3D model of the pen holder, detailing how the holder adjusts to different pen sizes.

- **3D Printing:**

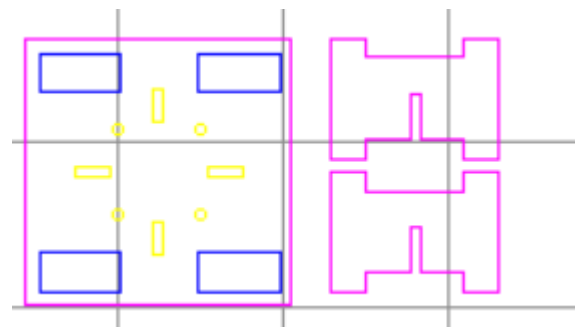
- After finalizing the design, the model was exported as an STL file for 3D printing. The pen holder was printed using (PLA) to ensure strength and durability during prolonged use.

d. Cart with Linear Bearings (Designed in Fusion 360, Cut in RDWorks)

The cart serves as the moving platform that carries the pen holder along the machine's axes. It was designed to work with 8mm linear bearings, which allow smooth motion along the shafts.

- **Design in Fusion 360:**

- The cart was designed in **Fusion 360** with precise slots for fitting the 8mm linear bearings. The size and placement of these slots were critical to ensure smooth movement without any wobbling.
- In the figure, the cart's design is shown with the attached bearings and pen holder, illustrating how it moves along the axis.



- **Conversion to RDWorks:**

- After completing the design, the cart was exported as a DXF file and imported into **RDWorks** for laser cutting.
- **Challenges:** The main challenge was ensuring the bearing slots were laser-cut with precise tolerances to avoid too much friction or slack, which could affect movement.

2. Assembly and Construction (2nd weeks):

- **Objective:** Build the CNC Plotter based on the design specifications.
- **Key Tasks:**
 - Assemble the frame, attach the gantry, lift arm, and timing belts.
 - Install stepper motors and pulleys.
 - Mount the control board and power supply.

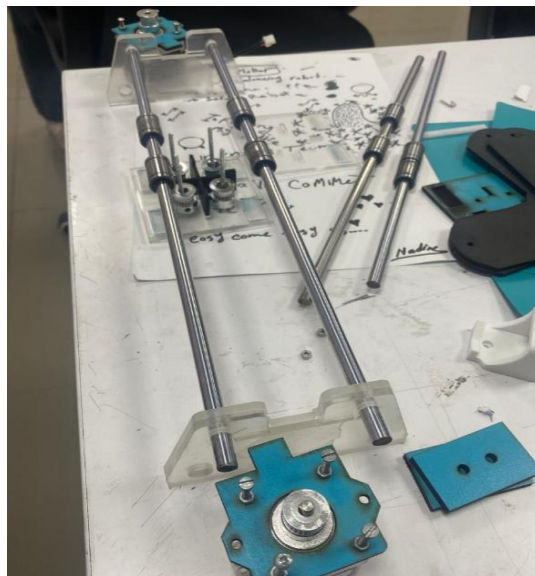
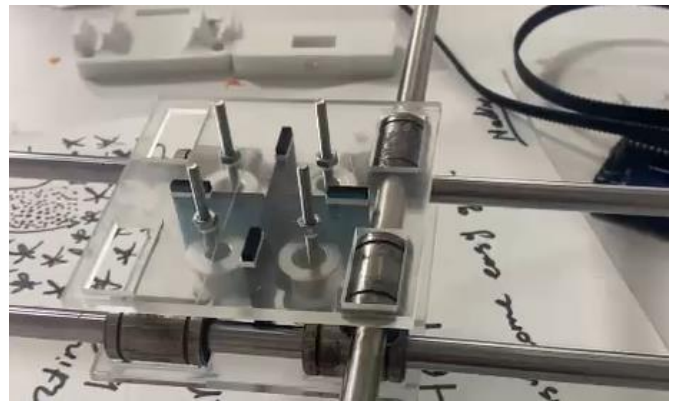
After laser cutting the parts in RDWorks and 3D printing the pen holder, the next stage was assembling the components.

a. Assembling the Axes and Shafts

- Mounted the laser-cut stepper motor supporters and shaft connectors.
- Installed the linear bearings and cart, making sure the movement was smooth.

b. Attaching the Pen Holder

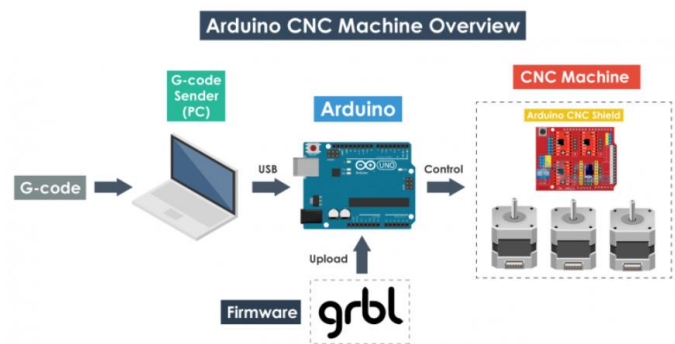
- The 3D-printed pen holder was attached to the cart, and adjustments were made for pen height.



3. Programming and Calibration (3rd weeks):

- **Objective:** Develop the code for controlling the CNC Plotter and calibrate the motors for precise movement.
- **Key Tasks:**
 - Write the software to convert digital designs into motor instructions.
 - Calibrate the X ,Y, and Z axes for accurate plotting.

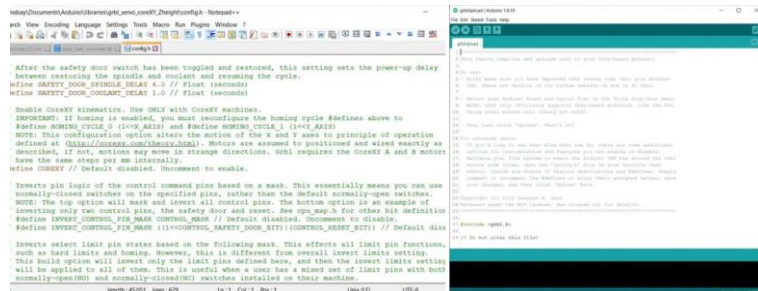
In this section, the process of programming and calibrating the CNC pen plotter will be described. GRBL software is used to control two stepper motors for the X and Y axes, as well as one servo motor for the Z-axis pen movement, all coordinated through G-code commands on an Arduino with a CNC shield and motor drivers.



3.1 GRBL Software

GRBL, an open-source firmware, is installed on the Arduino to interpret G-code and control the CNC hardware. This software is essential for controlling the plotter's stepper and servo motors.

- **GRBL Installation:**
 - GRBL is downloaded from GitHub and installed as a library in the Arduino IDE.
 - The GRBL firmware is then uploaded to the Arduino board (typically an Arduino Uno) by selecting the appropriate board and port.
- **Connecting GRBL to the CNC Shield:**
 - The CNC shield is attached to the Arduino board to facilitate the connection of motor drivers, stepper motors, and the servo motor.
 - The two stepper motors are connected to the X and Y axis terminals on the shield, while the servo motor is connected to one of the PWM pins for Z-axis control.



3.2 Stepper Motors for X and Y Axes

The two stepper motors are used to control the movement along the X and Y axes. These motors allow for precise movement in both clockwise and counterclockwise directions, enabling the plotter to move the pen across the drawing surface.

- **Clockwise and Counterclockwise Rotation:** Each stepper motor is controlled to rotate in either direction, adjusting the position along the X and Y axes.
- **G-code Commands:** Movement is commanded through G-code, with commands like *G01 X10 Y10* moving the pen to the designated coordinates.

Power is supplied to each stepper motor through a motor driver, such as the **A4988** or **DRV8825**, which amplifies control signals from the Arduino.

3.3 Servo Motor for Z-Axis (Pen Movement)

The Z-axis, which controls the vertical movement of the pen, is managed by a servo motor. This motor moves the pen up and down by rotating to specific angles, lifting or pressing the pen onto the paper.

- **Servo Motor Connection:** The servo motor is connected to a PWM pin on the CNC shield.
- **G-code Control:** G-code commands such as *M3* (to lower the pen) and *M5* (to raise the pen) are used to control the servo motor's position.

3.4 Calibration Process

Proper calibration is required to ensure that the plotter operates accurately along all axes and that the pen is correctly positioned during drawing.

- **Calibration of Stepper Motors:**
 - The GRBL settings are accessed using the *\$\$* command. The steps per millimeter for the X and Y axes (*\\$100* and *\\$101*) are adjusted so that the motors move the exact distances required.
- **Servo Motor Calibration:**
 - The Z-axis servo motor is calibrated to ensure the pen is lifted and lowered correctly. The angles for the pen-up and pen-down positions are set to ensure that the pen makes appropriate contact with the paper without excessive pressure.
- **Testing with G-code:**
 - A test pattern, such as a square or circle, is run using G-code to verify that the movements along the X, Y, and Z axes are accurate.



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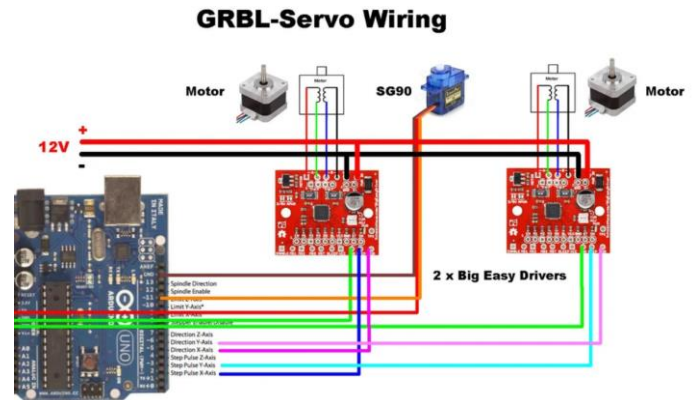
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$121=20.000
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$131=297.000
$132=20.000
ok
Autoscroll Show timestamp Newline 115200 baud Clear output

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3.5 Motor Driver Configuration

Motor drivers such as the A4988 or DRV8825 are used to control the current supplied to the stepper motors. The current must be correctly adjusted to match the specifications of the stepper motors to prevent overheating or underpowering.

- **Micro stepping Configuration:** Jumpers on the CNC shield are used to configure micro stepping, which increases the smoothness of motor movement by dividing each full step into smaller increments.
- **Current Adjustment:** The current limit on each driver is adjusted using a potentiometer to ensure the correct amount of current is delivered to the motors.



3.6 Plotting an Image

Once the CNC pen plotter is programmed and calibrated, the next step involves plotting an image or design using G-code. This process translates an image or drawing into a series of G-code commands that the CNC machine can execute, controlling the movement of the X, Y, and Z axes to recreate the image on the drawing surface.

3.6.1 Preparing the Image

Before plotting, the image must be converted into G-code. This can be done using various software tools that take a vector or raster image and generate the corresponding G-code file.

- **Image Conversion to G-code:**
 - **Inkscape** (with an extension like "Gcodetools") or **LaserGRBL** can be used to convert images into G-code.
 - Vector images (e.g., SVG format) work best, as they define shapes using lines and curves. Raster images (e.g., PNG or JPEG) may need to be traced or converted into vector format before generating G-code.
- **Setting Parameters:**
 - Ensure that the plot dimensions (X and Y axis limits) in the G-code match the physical dimensions of the CNC plotter's working area.
 - Set the pen up and pen down heights correctly for the Z-axis to avoid damaging the paper or hardware.

3.6.2 Loading the G-code File

Once the image is converted into G-code, the file is uploaded to the GRBL controller software, which will send the commands to the Arduino controlling the CNC machine.

- **Use GRBL Controller Software:**
 - Open a GRBL controller program like **Universal Gcode Sender (UGS)** or **bCNC**.
 - Connect the Arduino to your computer and load the G-code file.
 - Position the pen at the origin (home position) of the drawing area, ensuring the pen is above the surface.
- **Start the Plotting Process:**
 - Initiate the plotting process by sending the G-code commands to the CNC plotter. The plotter will follow the paths described in the G-code, moving the X and Y stepper motors to trace the image on the paper, while the Z-axis servo motor raises and lowers the pen as needed.

3.6.3 Monitoring the Plotting Process

It is important to monitor the plotter during the drawing process to ensure smooth operation:

- **Check Motor Movements:** Ensure that the X and Y stepper motors are moving correctly and that the pen is following the designated paths.
- **Pen Positioning:** Verify that the Z-axis servo is properly lifting and lowering the pen at the correct points to avoid unintended marks or smudges on the paper.
- **Pause and Resume:** If adjustments are necessary, most GRBL controller software provides options to pause and resume the process.

3.6.4 Post-Plotting Adjustments

Once the image has been plotted:

- **Check Accuracy:** Review the plotted image for accuracy. If adjustments are needed, recalibrate the motors or adjust the G-code parameters.

4. Testing

- **Objective:** Test the CNC Plotter to ensure accurate and consistent operation.
- **Key Tasks:**
 - Test various designs and adjust motor speeds or timing belt tension if necessary.
 - Fine-tune the system for repeatable precision.

a. Square Drawing Test

- **Purpose:** The test aimed to check if the CNC machine could accurately reproduce a basic geometric shape (a square), with all sides being equal and all corners forming right angles.
- **Procedure:**
 1. **Step 1:** A command was sent through the GRBL software to instruct the machine to draw a square with predefined dimensions (e.g., 5 cm by 5cm).
 2. **Step 2:** The stepper motors were observed to ensure smooth, coordinated movements of the X and Y axes, producing straight lines and sharp 90° turns at each corner of the square.
 3. **Step 3:** The servo motor controlling the pen holder was activated at the appropriate times, ensuring the pen was lowered for drawing and raised when repositioning between moves.
- **Expected Outcome:** The drawn square was expected to have four straight sides of equal length, with precise 90-degree angles at each corner. Any deviations from the expected results would indicate issues with calibration or alignment.

b. Results:

- The machine successfully drew a square, and the dimensions of the shape were measured using a ruler to verify the accuracy of the plotter's movements.
- The results showed minimal deviation in the length of the sides, confirming the proper functioning of the X and Y axis motors.
- The servo motor effectively controlled the pen, ensuring smooth transitions between pen-down and pen-up positions, without dragging the pen.

c. Adjustments:

- Minor adjustments were made to the stepper motor calibration to fine-tune the precision, resulting in even more accurate subsequent tests.

The **square drawing test** served as a fundamental benchmark for evaluating the CNC plotter's basic performance and identifying areas for improvement before more complex patterns were attempted.

To view the video demonstration of the **square drawing test**, please follow this link: [CNC Plotter - Square Drawing Test Plan Video](#)

This video captures the machine's movements during the square test, providing a visual reference for the test plan phase.

5. Conclusion

The CNC machine plotter project was successfully completed, with all components designed, fabricated, and assembled according to plan. Throughout the process, the stepper motor supporters, shaft connectors, cart with linear bearings, and 3D-printed pen holder were developed using precise laser cutting and 3D printing techniques.

The programming and electronics, including the CNC shield, GRBL software, Arduino, and servo motor, were integrated seamlessly to control the machine's movements. The plotter underwent rigorous testing, where it demonstrated accurate and reliable operation. Minor adjustments were made during testing to fine-tune the machine's performance.

In conclusion, the CNC plotter has met its design objectives, providing precise drawing capabilities and smooth movement across both the X and Y axes. The machine is now fully operational and ready for practical use.